

Physics-Informed DMD (piDMD)

Aim: Learn linear models from snapshot data.

Recap Given snapshot matrices

$$X = \begin{bmatrix} | & & | \\ x_1 & \dots & x_m \\ | & & | \end{bmatrix} \quad \text{and} \quad Y = \begin{bmatrix} | & & | \\ y_1 & \dots & y_m \\ | & & | \end{bmatrix},$$

$d \times m$ $d \times m$

where $y_j = \tau x_j \quad j=1, \dots, m,$

DMD constructs the best linear fit

$$(*) \quad \hat{\tau} = \underset{S \in \mathbb{R}^{d \times d}}{\operatorname{argmin}} \|Y - SX\|_F.$$

$\Rightarrow$ Standard solution is $S = YX^+$.

$\Rightarrow$ Compute eigendecomposition of S
w/out explicitly forming $d \times d$ matrix:

$$X^+ Y w = \lambda w \quad \lambda \neq 0 \quad \Leftrightarrow \quad S v = \lambda v, \quad v = Y w$$

$\Rightarrow$ For stability, use SVD $X = U \Sigma V^T$
and truncate to rank $r \leq m$.

DMD Algorithm (truncated to rank r)

Step 1. Compute $X_r \approx U_r \Sigma_r V_r^T$
r-truncated SVD

Step 2. Solve $(U_r^T Y V_r \Sigma_r^+) w = \lambda w$

Step 3. Return eigenvalues $\{\lambda_k\}_{k=1}^r$ and
eigenvectors $\{Y V_r \Sigma_r^+ w_k\}_{k=1}^r$ of the
matrix $\tilde{T}_r = U_r^T \hat{T} U_r$.

Note: The matrix $\tilde{T}_r$ is called the compression
of the matrix $\hat{T}$ to the subspace $\text{col}(U_r)$.
It generalizes the Rayleigh Quotient of $\hat{T}$.

Note: If $r=m$, then the eigenvectors/values
of $\tilde{T}_r$ and $\hat{T}$ are the same!

Fundamental Limitations of DMD

DMD has many **useful features**:

⇒ Simple to implement and apply.

⇒ Identifies dynamically relevant features that are well-represented in the snapshot data.

⇒ May provide good short-term predictions, e.g., for applications in controlled systems.

However, it also has **fundamental limitations**:

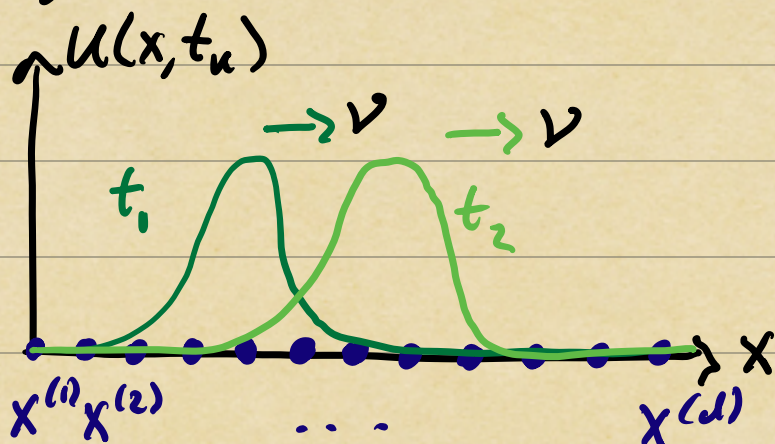
⇒ Often fails to generalize beyond snapshot data, since predictions $\in \text{col}(Y)$.

⇒ Underlying dynamics may not be well-represented by low-rank matrix.

⇒ May fail to capture nonlinear effects.

Example. 1D Advection.

Suppose that we have snapshot data from a Gaussian pulse traveling to the right at a constant velocity $v > 0$.



$$X = \begin{bmatrix} | & & | \\ x_1 & \dots & x_m \\ | & & | \end{bmatrix}$$

Q: If $m \ll d$, can DMD find a "good" linear model for the advection problem?

$\Rightarrow$ All predictions will be linear combos of the observed wave profile. But if $m \ll d$, we will not necessarily be able to form all translates of the wave profile from the snapshots.

$\Rightarrow$ In other words, translation is not low-rank!

Q: What structure does the true model have?

For simplicity, suppose the wave speed is

$$v = \frac{\Delta x}{\Delta t},$$

so that Δt is the time for the wave to shift to the right by $\Delta x = \text{one grid point}$.

$$\begin{bmatrix} u(x^{(1)}, t_{k+1}) \\ \vdots \\ u(x^{(d)}, t_{k+1}) \end{bmatrix} = T \begin{bmatrix} u(x^{(1)}, t_k) \\ \vdots \\ u(x^{(d)}, t_k) \end{bmatrix} = \begin{bmatrix} 0 \\ u(x^{(1)}, t_k) \\ \vdots \\ u(x^{(d-1)}, t_k) \end{bmatrix}$$

$$\Rightarrow T = \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ & 1 & 0 & \\ & & \ddots & 0 \\ & & & 1 \end{bmatrix} \quad \text{"rightward shift"}$$

$\Rightarrow$ The singular values of T are

$$G_k = \begin{cases} 1 & 1 \leq k \leq d-1 \\ 0 & k=d \end{cases}$$

So any rank- m approx. w/ $m \leq d$ will

have large errors. By Eckart-Young,

$$\min_{\text{rank}(B)=m} \|T-B\|_2 \geq \sigma_{m+1} = 1 \quad (m+1 \leq d),$$

or in the Frobenius norm,

$$\min_{\text{rank}(B)=m} \|T-B\|_F \geq \sqrt{\sum_{j>m} \sigma_j^2} = \sqrt{d-m}.$$